

33-Seaborn Categorical Data Analysis

Seaborn Categorical Data Analysis – Lecture Summary

Introduction

- The lecture focuses on categorical data analysis using the Seaborn Python library.
- The session aims to demonstrate practical plotting and interpretation of categorical data structures in pandas DataFrames.

Loading Categorical Data

- The 'tips' dataset from Seaborn's built-in datasets is used.
- Key dataset columns: total bill, tip, sex, smoker, day, time, size.
- Some columns are inherently categorical (e.g., day, sex, time), while others may appear numeric but are categorical due to a fixed set of values (e.g., size).

Types of Plots in Seaborn

- **Strip Plot**
: Shows the distribution of a categorical value versus a numerical variable. E.g., tip amount across four days (Thursday to Sunday). Distribution is not always fully clear with strip plots.
- **Swarm Plot**: Offers a clearer view of point distributions, e.g., exact tip values given on each day.

- **Box Plot**

: Displays five-point summary statistics (median, interquartile ranges, whiskers, outliers). Used for statistical analysis and to understand spread and outliers.

- **Violin Plot**

: Combines KDE plots to visualize distribution and density. Shows whiskers, interquartile ranges, and median.

- **Bar Plot**

: Aggregates values over categories. Similar to histogram, but specifically for categorical data. Features error bars, which indicate potential variance/discrepancy.

- **Count Plot**

: Counts instances in each category (e.g., number of male vs female customers). Y-axis shows counts by default.

- **Categorical Plot (catplot)**

: A versatile function that can render various categorical plot types (e.g., count, box) using the 'kind' attribute.

Plot Interpretations & Attributes

- For most categorical plots, either the x or y attribute must be categorical, not both.
- The 'hue' parameter allows additional categorical breakdown (e.g., group sizes within days).
- Elements explained: whiskers, interquartile ranges, median (for box/violin); error bars (for bar plot).
- Differences between plots: Each provides unique insights—distribution (swarm/strip), summary statistics (box/violin), aggregate comparisons (bar), category counts (count plot).

Practical Demonstration & Student Q/A

- Live code demonstrations included dataset loading and different plot generations.

- Questions addressed: categorical vs numerical distinction, function parameters (x/y/data), application of string/categorical conversion, and plot selection criteria.

Upcoming Classes & Exam Information

- Lecturer confirms completion of Python material, with one more class for revision/practice, covering missing methods, especially in NumPy/Pandas.
- The upcoming exam (Python test) is scheduled for Friday at 8 a.m.; it will be easy and largely MCQ-focused, especially if assignments have been completed.
- Next tests (e.g., Statistics) will be tougher.
- Students can expect further practice and revision sessions.

Conclusion

- The lecture concludes with reassurance about the ease of the test and the comprehensive coverage of Python material, including Seaborn's use for categorical data visualization.
- Students are encouraged to prepare and revise, with an additional session planned for review before the exam.

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